



## **BANNARI AMMAN INSTITUTE OF TECHNOLOGY**

An Autonomous Institution Affiliated to Anna University – Chennai • Approved by AICTE • Accredited by NAAC with "A" Grade

**SATHYAMANGALAM - 638401 ERODE DISTRICT TAMILNADU INDIA**

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## **22GE004 – BASICS OF ELECTRONICS ENGINEERING**

### **UNIT I & ENERGY TRANSFER AND SIGNALS**

#### **1. ENERGY TRANSMISSION THROUGH MECHANICAL, ELECTRICAL AND ELECTROMAGNETIC MEANS, SIGNAL AS ENERGY TRANSMISSION, COMPLEXITY IN SIGNAL TRANSMISSION**

Energy transmission is the process of transferring any signal from one place to another. It takes place with the help of any of the three means.

1. Energy transmission through mechanical mean
2. Energy transmission through electrical mean
3. Energy transmission through electromagnetic mean

##### **1.1. Energy transmission through mechanical mean**

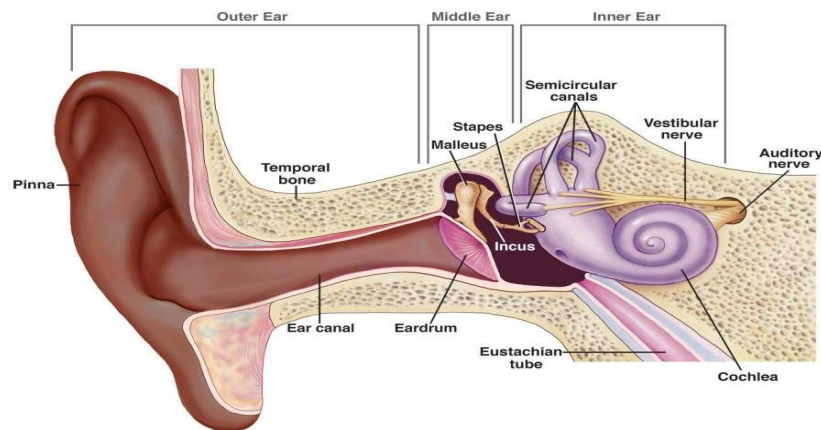
Energy transmission through mechanical means happens without the help of any medium or equipment. we can easily understand with the help of the following scenario.

**Scenario: Hearing a sound signal using the ears.**

A. Sound waves enter the outer ear and travel through a narrow passageway called the

ear canal, which leads to the eardrum.

- B. The eardrum vibrates from the incoming sound waves and sends the vibrations to 3 tiny bones in the middle ear. These bones are called the malleus, incus, and stapes.
- C. The bones in the middle ear amplify, or increase, the sound vibrations and send them to the cochlea, a snail-shaped structure filled with fluid, in the inner ear. An elastic partition runs from the beginning to the end of the cochlea, splitting it into an upper and lower part. This partition is called the basilar membrane because it serves as the base, or ground floor, on which key hearing structures sit.
- D. Once the vibrations cause the fluid inside the cochlea to ripple, a traveling wave forms along the basilar membrane. Hair cells—sensory cells sitting on top of the basilar membrane—ride the wave. Hair cells near the wide end of the snail-shaped cochlea detect higher-pitched sounds, such as an infant crying. Those closer to the center detect lower-pitched sounds, such as a large dog barking.
- E. As the hair cells move up and down, microscopic hair-like projections at perch on top of the hair cells bump against an overlying structure and bend. Bending causes pore-like channels, which are at the tips of the stereocilia, to open up. When that happens, chemicals rush into the cells, creating an electrical signal.
- F. The auditory nerve carries this electrical signal to the brain, which turns it into a sound that we recognize and understand.



**Figure 1. Parts of the ear**

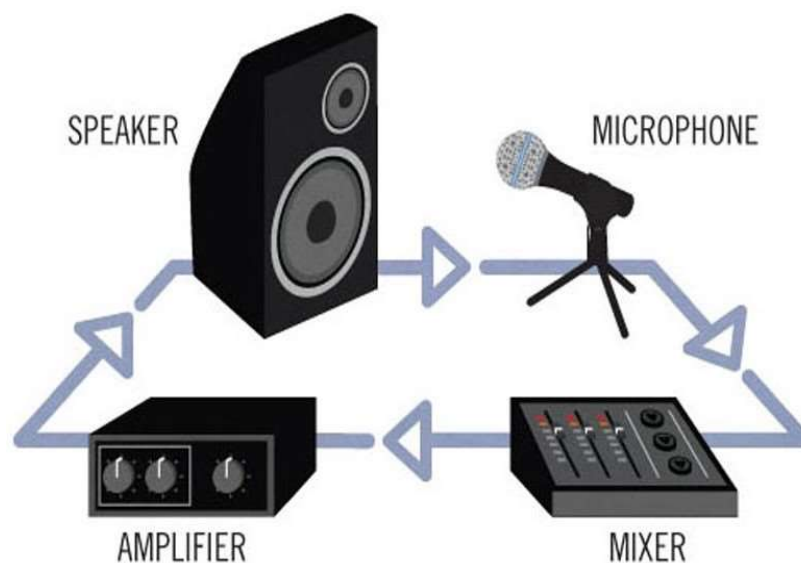
Video link: <https://www.youtube.com/watch?v=eQEaiZ2j9oc>

## 1.2. Energy transmission through electrical mean

Energy transmission through electrical means happens using a sensor and an actuator to send a signal from one place to a relatively larger place than a mechanical mean. we can easily understand with the help of the following scenario.

### **Scenario: Send sound signal using microphone and speaker.**

The ability to translate sound into electrical signals and back again is truly remarkable. Whether you're talking on the phone, playing your favorite music in your car, creating music for your next album, or watching a movie on your laptop, you're using microphones and speakers to make it all happen. Both mics and speakers are necessities in a home studio. However, their applications are not limited here – you can find mics and speakers integrated into a single device, such as gaming headphones and police radios. Microphones and speakers work together to record and play music, although separate technologies go into the development of each device. Microphones capture sound waves from different directions (based on the polar pattern and convert them into electrical signals. Speakers then take those electrical signals and turn them back into sound waves that an audience can hear.



**Figure 2. Microphone and Speaker**

## **How Microphones Work**

A microphone is a transducer that converts sound energy into electrical energy. Microphones are considered a vital determinant of the resulting sound quality. If your mic is poor quality, worn out, and damaged to the extent of producing harsh or distorted signals, that's what you'll get in the end speaker output. Audio interfaces and DAW software can help eliminate some noise, but this can be difficult to achieve. As such, it is important to focus on getting a clean signal from your microphone in the first place. To understand how a microphone works, we need to understand its various components. An average microphone consists of the following parts.

- A voice coil: Often made of copper or aluminum wire wound around a bobbin. When vibrated, it creates disturbances in the magnetic field.
- A permanent magnet: Creates a permanent magnetic field around the voice coil.
- A diaphragm: Receives sound and vibrates in a manner that corresponds to the speaker's voice.
- Signal-carrying wires or transmitter: The electrical signals created by the movement of the voice coil around a permanent magnet are transmitted to the next processing phase by wires or sent to a receiver by a wireless transmitter.
- Other connections: Modern microphones offer a wide variety of connectors that allow you to connect your device to an amplifier, receiver, or computer. USB is one of the most popular connectors for home studio mics since it allows you to connect directly to the computer.

When you speak directly into the microphone, your voice changes the sound pressure, which moves the diaphragm attached to the voice coil. As the voice coil vibrates inside the magnetic field of the permanent magnet, it creates an electrical signal that matches your voice. The signal generated will vary according to the mic's component quality. The signal-carrying wires (usually XLR cables) carry the resulting signal to the amplification stage. Some microphones have a wireless transmitter that sends these signals wirelessly to a receiver located a couple of feet within the room. That said, all microphones sound different due to their size and quality differences.



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Different microphones contain different transducer technologies, giving your mic unique sound properties. Also, the speaker's unique voice contributes to the differences in microphones.

### **How Speakers Work**

Like microphones, speakers are transducers, but in this case, they convert electrical energy into sound energy. Speakers take in an amplified signal and pass it through a voice coil to create sound.

To better understand how speakers work, let's dive into the basic structure of a speaker:

- Speaker cone: Vibrates to create sound.
- Voice coil: The electromagnet attached to the diaphragm.
- Spider: Flexible mechanism that restores the voice coil to its original position between movements.
- Dust cap: Located at the center of the diaphragm, protecting it from dust.
- Permanent magnet: Creates a permanent magnetic field around the voice coil.
- Enclosure: Every speaker comes in an enclosure made of wood or plastic to prevent sound loss from wave cancellation.
- First, the mic-level signal goes into a preamp, where a series of transistors and vacuum tubes raise its voltage to a line-level signal. Preamps also add noise and distortion to the signal.
- The now powerful line-level signal travels to an audio interface, where it's fed into your computer for processing. In this stage, you can add EQ, distortion, reverb, and echo through a digital audio workstation or other editing software.
- The edited soundtrack can be channeled to an amplifier through a quarter-inch (6.35 mm) jack. Unfortunately, most audio interfaces do not have an integrated amplifier. You need an external amplifier to boost the signal to a more powerful voltage that a speaker can play.



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- The analog signal goes through your speaker's voice coil, which turns into an electromagnet. The electromagnet moves back and forth within the permanent magnet surrounding it. Since the voice coil is attached to a diaphragm, both move at the same velocity, pushing the air around the diaphragm and creating sound.

As you may have guessed, several other components come into play before the sound gets to the speaker cone and is disbursed to the audience. Some of these components include:

- Preamplifier
- Amplifier
- Audio interface
- DAW and other editing software

### **Preamplifier**

The mic signal is one of the weakest signals. It cannot drive a speaker on its own without being amplified. Microphone preamps can exist as standalone units, although they usually exist as elements of a much larger system, e.g., a mixer.

### **Amplifier**

An amplifier's main duty is to convert a line-level signal to a speaker-level signal or the more powerful signal that can actually drive a speaker. Amplifiers come in all shapes and sizes, with the most powerful ones putting out over 2000 watts per channel. Amplifiers can also exist as standalone units (external amps), while others can be part of your receiver or mixer. Other types of amplifiers can also exist as software that can be downloaded and used on your computer.

### **Audio Interface**

An audio interface is another must-have piece of equipment in your studio setup. Audio interfaces provide all the ports you need to hook up your hardware and instruments to your computer. However, its most important role is converting input into a format your computer can understand. A digital-to-analog converter or DAC helps in this task. A digital-to-analog converter is a device that takes the soundtrack that's processed and stored in binary in your





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computer and converts them into an analog waveform that a speaker can play. Audio interfaces are said to have much more processing power than your computer's internal sound card.

Video Link: <https://www.youtube.com/watch?v=ZX8QwGKaOG8>

### **1.3. Energy transmission through electromagnetic mean**

Energy transmission through electromagnetic means happens while trying to send a signal to larger distances. we can easily understand with the help of the following scenario.

#### **Scenario: Mobile network**

##### **Cell Phones use More Advanced Technology than Landlines:**

As telephone technology has grown over the years, we are now in the era of smartphones. In general, much of our daily communications rely on WiFi and the use of the internet. Smartphones are more than just a phone!

In a similar way to a landline, they allow two (or more) people to converse. However, they also integrate much more advanced technology that allows users to text, video call, and access the internet.

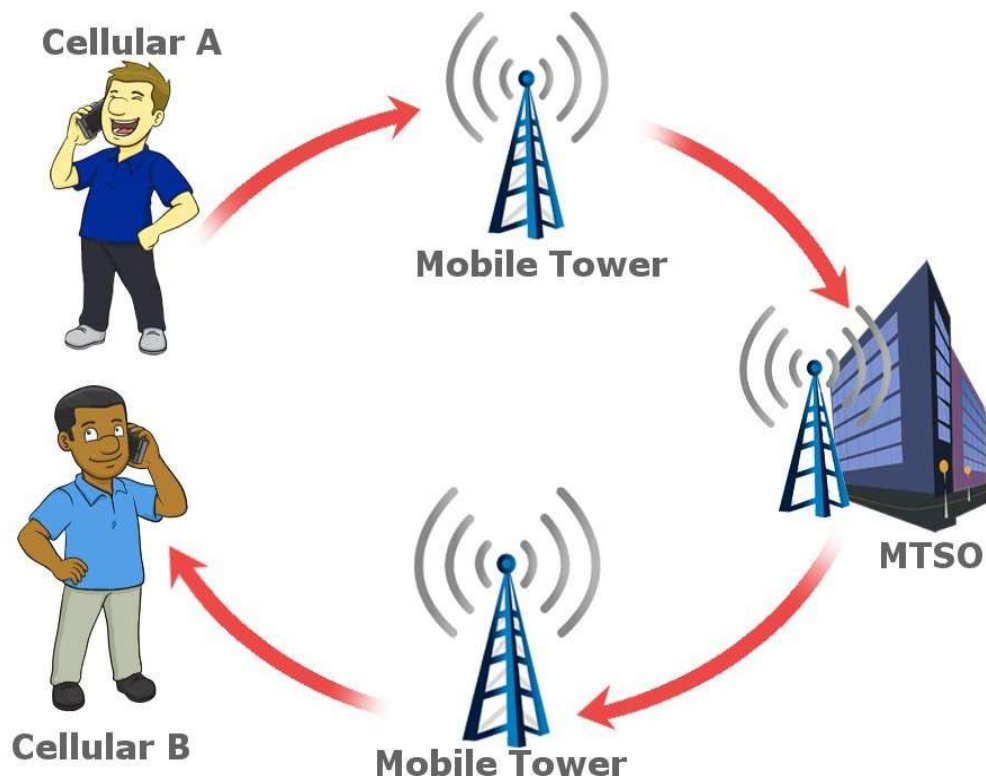
##### **Cell Phones use Electromagnetic Radio Waves:**

While landlines use electrical currents to send the sound of your voice to the receiver, cell phones use wireless technology. Cell phones use only electromagnetic radio waves to send and receive the same sound that would otherwise have to travel down wires. With the increase of all kinds of technology, whether it's televisions, radios, phones, or any other wireless device (even small things such as a wireless doorbell), we cannot escape electromagnetic waves. No matter where you go, they are all around us! According to Explain That Stuff, cell phone networks are the fastest-growing source of electromagnetic energy that exists.

### How Wireless Technology Works:

But how does your voice wirelessly get to your co-workers, friends, and family? Similarly to landlines, when you speak into your cell phone, a microphone within picks up on your voice – and all of its tone variations. It then converts these vibrations into electrical signals (just like a landline!)

Now, here is where things are a bit different from the way a landline works. Once inside the phone, the signals are picked up by a microchip. This microchip converts the signals into strings of numbers. Then, these numbers are ‘packed up’ into a radio wave, and then released by your phone’s antenna. Once the radio wave is released, it rushes through the air until it finds a cell phone mast.



**Figure 3. Mobile network**





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Once the mast gets the radio wave, it sends it to the ‘base station.’ Explain That Stuff describes what happens once the radio waves hit the base station: “From the base station, the calls are routed onward to their destination. Calls made from a cell phone to another cell phone on the same network travel to their destination by being routed to the base station nearest to the destination phone, and finally to that phone itself.” As you can see, both landlines and cell phones rely on electronic waves to send and receive signals. However, the way that these electronic waves are sent and processed varies between wireless and wired connections.

Video link: [https://www.youtube.com/watch?v=1JZG9x\\_VOwA](https://www.youtube.com/watch?v=1JZG9x_VOwA)